

Chocolate at heart: the anti-inflammatory impact of cocoa flavanols.

Chronic and acute inflammation underlies the molecular basis of atherosclerosis. Cocoa-based products are among the richest functional foods based upon flavanols and their influence on the inflammatory pathway, as demonstrated by several *in vitro* or *ex vivo* studies. Indeed, flavanols modify the production of pro-inflammatory cytokines, the synthesis of eicosanoids, the activation of platelets, and nitric oxide-mediated mechanisms. A relative paucity of data still characterizes the *in vivo* implications of these findings albeit there have been studies suggesting that the regular or occasional consumption of cocoa-rich compounds exerts beneficial effects on blood pressure, insulin resistance, vascular damage, and oxidative stress. Accordingly, rigorous controlled human studies with adequate follow-up and with the use of critical dietary questionnaires are needed to determine the effects of flavanols on the major endpoints of cardiovascular health.